HW2 Assignment 3 Competitive Market. Due Date: Nov 8/2026

Competitive Market framework ⇒ A firm is price taker its accept market equilibrium price & choose its output where $P = MR = MC$

1) Market equilibrium price & quantity.
2) Output produced by a typical firm.
3) Profit or loss earned by typical firm.

Part a) Step1

$P_{demand} = 40 - 0.25Q$

Market supply $P_s = 5 + 0.05Q$

(Step3) Calculate Equilibrium Price

$Q = 116.66$ substitute into either

Demand or Supply $P_s = 5 + 0.05(116.66)$

$P = 5 + 5.833$

$= 10.83\ \$$

Market Equilibrium (step2)

$Q_D = Q_S$

$40 - 0.25Q = 5 + 0.05Q$

$35 = 0.3Q$

$Q^* = \frac{35}{0.30} = 116.66$

Market Equilibrium output = 116.66 units

Market Equilibrium price = $10.83

Part b Determine output for Typical firm.

$TC = 100 - 20q + 2q^2$

$MC = \frac{\partial TC}{\partial Q} = -20 + 4q$ → Π_{max} condition

$MR = MC$

$P = MR = MC$

$10.83 = -20 + 4q$

$10.83 + 20 = 4q$

$30.83 = 4q$

$q^* = \frac{30.83}{4} = 7.71$ units profit maximized.

$P = 10.83$

$MR = 10.83$

Part C Determine Profit or Loss

$\Pi = TR - TC$

$TR = P \times Q$

$= 10.83 \times 7.71$

$TR = 83.49\ \$$

$TC = 100 - 20(7.71) + 2(7.71)^2$

$TC = 64.67\ \$$

$\Pi = TR - TC = 83.49\ \$ - 64.67\ \$ = 18.84\ \$$

→ Firm earns positive economic profit

Question 2

Given # of firms = 100 short run total cost ⇒ $STC = 0.5q^2 + 10q + 5$

$Q_D = 1100 - 50P$ market output Q is measured in total snuff box per day

Step1 Derive cost functions $\{TC = TFC + TVC\}$

① $TFC = 5$ ② $TVC = 0.5q^2 + 10q$

③ $MC = \frac{\partial STC}{\partial Q} = q + 10$ ④ $AVC = \frac{TVC}{q} = \frac{0.5q^2 + 10q}{q} = \frac{q(0.5q+10)}{q}$

$= 0.5q + 10$

⑤ $ATC = \frac{STC}{q} = 0.5q + 10 + \frac{5}{q}$ ⇒ These 5 cost concepts will help solve entire question.

Part a

Firm max profit where $P = MR = MC$

$MC = q + 10$ since $P = MC$ then $P = q + 10$

$Q = P - 10$

min AVC for shut down condition

$AVC = 0.5q + 10$

min $q = 0$ min $AVC = 10$

If $P < 10$ ⇒ firm shuts down

$P = 10$ ⇒ indifferent

$P > 10$ ⇒ Produce according to its MC curve.

$$q_s = \begin{cases} 0 & P \le 10 \\ P - 10 & P > 10 \end{cases}$$ The individual short-run supply function.

Part b

Firms = 100

Each firm's supply $q = P - 10$ when $P > 10$

Market supply is the horizontal of all individual supply Q.

$Q_S = 100(P - 10) = 100P - 1000$ ⇒ rearrange inverse supply function

Individual supply

$Q = P - 10$

Market supply

$Q = 100(P - 10)$

$Q_S = 100P - 1000$

$\frac{Q + 1000}{100} = \frac{100P}{100}$

$P = 10 + 0.01Q$

$$Q_S = \begin{cases} 0 & P \le 10 \\ 100P - 1000 & P > 10 \end{cases}$$

Part c Determine market equilibrium and each firm's total short-run Profit?

Market Demand $Q_D = 1,100 - 50P$

Market Supply $Q_S = 100P - 1000$

Step 1

$Q_D = Q_S$

$1,100 - 50P = 100P - 1000$

$2,100 = 100P + 50P$

$2,100 = 150P$

$P = \frac{2,100}{150} = 14\ \$$

Step 2

Find Equilibrium Q_{Market}

Subst. $P = 14$ into Demand.

$Q_D = 1,100 - 50(14) = 400$

$Q^*_D = 400$

step 3 Find output for each firm

$q = \frac{Q}{N} = \frac{400}{100} = 4$

$q^* = 4$

Individual supply function (Alternative way)

$Q = P - 10 = 14 - 10 = 4$

Step 4 Calculate each firm's total short-run profit.

Economic Profit $= TR - TC$

$TR = P \times Q = 14 \times 4 = 56$

$TC = 0.5q^2 + 10q + 5$.

$q = 4 \Rightarrow TC = 0.5(4)^2 + 10(4) + 5 = 53$

$\Pi = TR - TC$

$= 56 - 53$

$\Pi = 3\ \$$ per firm per day.

Since there 100 firms

In dustry Profit $= 3 \times 100 = 300\ \$$

Part D — Market Equilibrium & Producer surplus graph.

$Q_D = 1{,}100 - 50P$ (Market Demand)

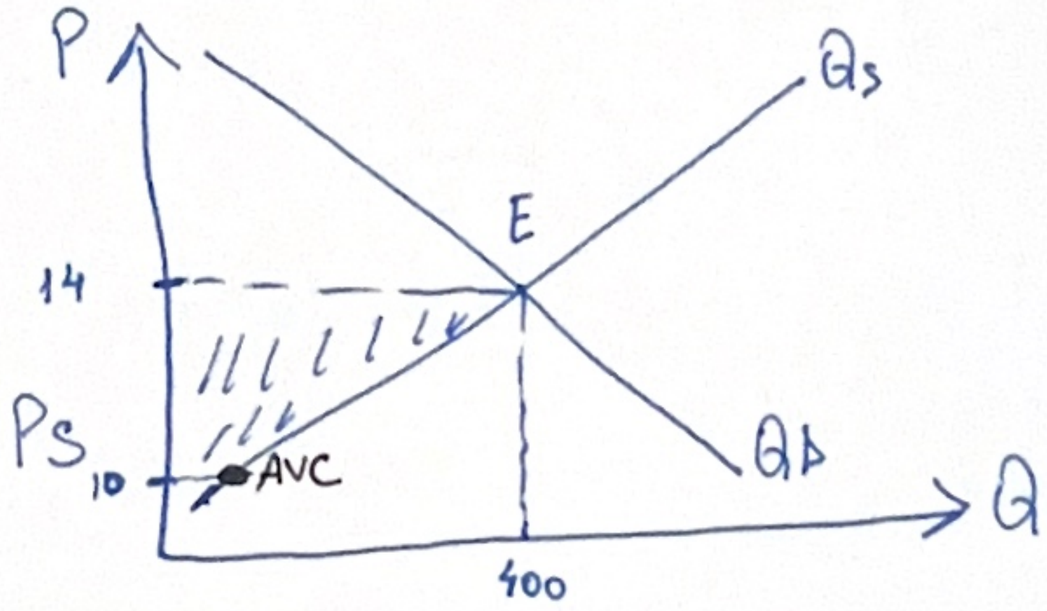
inverse demand $P = \frac{1100}{50} - \frac{Q}{50} = 22 - 0.02Q$

Market Supply

$Q_S = 100P - 1000$

inverse supply $P = \frac{1000}{100} + \frac{Q}{100} = 10 + 0.01Q$

$E(Q^*, P^*) = (400, 14)$



$PS = \frac{1}{2} \times \text{base} \times \text{height}$ (base: Equilibrium)

$= \frac{1}{2} \times 400 \times (14 - 10)$

$= 200 \times 4 = 800$

PS = 800 $ total market producer surplus per day.

PS = Economic Profit + Fixed cost (≥ 5)

$= 300 + (100 \times 5)$

$= 800\$$

of firms = 100

Part e

A tax on producers increase their effective MC by the amount of tax

Original MC = $q + 10$ — tax = 3$

$MC_{tax} = MC = q + 10 + 3 = q + 13$

$P_{Consumer} = MC + Tax$

$= q + 13$

Firm receives consumer price minus the tax

$P_S = P_C - 3$ $ ⟹ For 100 firms $Q_S = 100(P_C - 13)$

$= 100P_C - 1300$

New Equilibrium Price $Q_D = Q_{S\,New}$

$1{,}100 - 50P_C = 100P_C - 1300$

$1{,}100 + 1300 = 150P_C$

$P_C = \frac{2400}{150} = 16\$$ → Price paid by Consumer.

Parte

step3/ Calculate new equilibrium quantity

$$Q_D = 1,100 - 50(16)$$

$$= 1,100 - 800$$

$$Q = 300$$

step4 Price received by producers

$$P_p = P_c - Tax$$

$$P_p = 16 - 3 = 13\$$$

	before tax	after tax
CS	14 $	16 $
PS	14 $	13 $
Market Q	400	300
Output per firm	4	3
Tax per unit	0	3 $

step5/ Calculate output per firm.

$$q = \frac{Q_D}{N} = \frac{300}{100} = 3$$

$$TR_{tax} = Tax \times Q$$

$$= 3 \times 300$$

$$= 900$$

Government collects tax revenue.

Part 1

$PS_{before} = 800$

Loss D/ PS $= 800 - 450 = 350\$$

$$PS_{after} = \frac{1}{2} \times 300 \times (13-10) = 450$$

Before Taxation

$q = 4 \quad P = 14$

$TR = 14(4) = 56$

$TC_{before} = 53$

$\Pi_{before} = 56 - 53$

$= 3$ per firm

$\Pi_{before\ 100\ firms} = 300$

$\Delta\Pi = 300 - (50) = 350$

After taxation

$q = 3 \quad P_p = 13$

$TR = 13 \times 3 = 39$

$TC = 39.5$

$\Pi_{after} = 39 - 39.5 = -0.5$

$\Pi_{after} = -0.5\$$ per firm

$\Pi_{after\ 100\ firms} = 100 \times (-0.5) = -50\$$

Part f Why don't fixed enter into this computation of change in SR producer surplus?

$PS = TR - TVC$

Economic Profit

$\pi = TR - TVC - TFC$

The relationship $PS = \pi + TFC$

in short TFC do not change

$TFC = 100(5) = 500$

Total industry cost = 500 ₱

before taxation

$PS = 300 + 500 = 800$

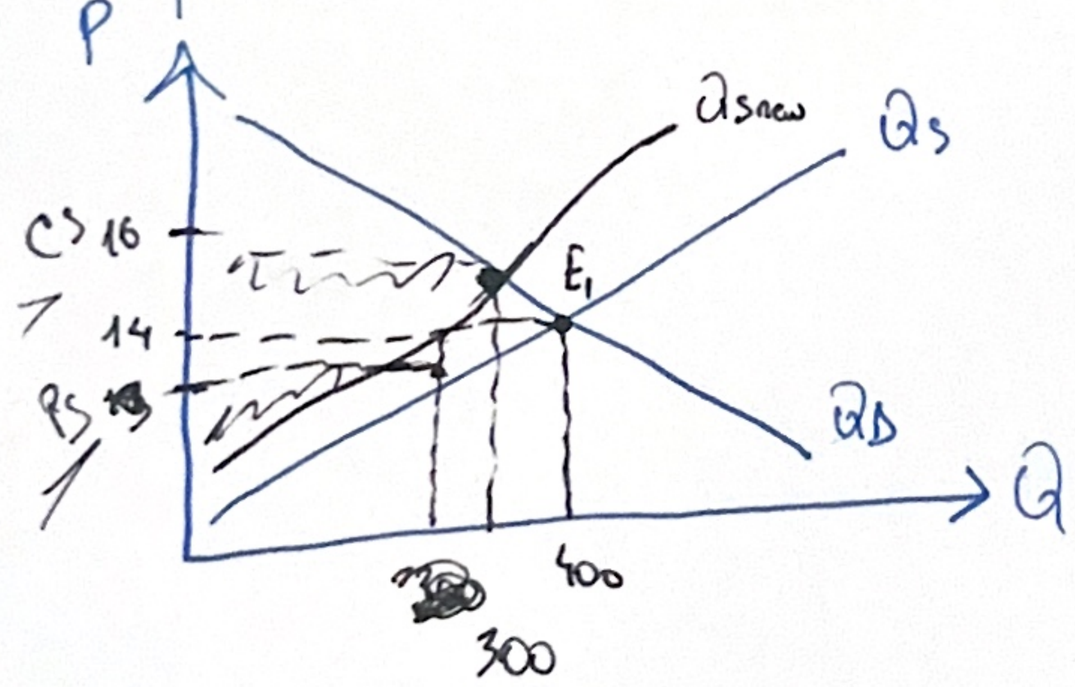
After taxation

$PS = -50 + 500 = 450$

$\Delta PS = 800 - 450 = 350$

$\Delta \pi = 300 - (-50) = 350$

TFC remain constant regardless of the firm's output decision in the SR.



$\Delta WL = \frac{1}{2}(3)(400 - 300)$

$\Delta WL = 150$ ₱

HW 7 A Shut down Case

loss > AFC

The firm should shut down the operation